



Set Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Dictionaries.

Section 1: Set Basics and Creation

★ Create and Display a Set

★ Create a Set from a List (Remove Duplicates)

Check Membership using in Operator

★ Find Length of a Set

Iterate through a Set using Loop

Empty Set Creation using set()

Section 2: Adding and Removing Elements

★ Add a Single Element to a Set

★ Add Multiple Elements using update()

★ Remove an Element using remove()

Remove an Element using discard()

★ Remove a Random Element using pop()

Clear All Elements from a Set

Section 3: Set Operations

★ Union of Two Sets

★ Intersection of Two Sets

★ Difference of Two Sets

Symmetric Difference of Two Sets

★ Check if a Set is a Subset

Check if a Set is a Superset

Check if Two Sets are Disjoint

Section 4: Set Comprehension and Conversion

★ Create a Set using Set Comprehension

Filter Even Numbers using Set Comprehension

★ Convert Set to List and Sort it

Convert String to a Set of Characters



Frozenset Creation and Usage

Section 5: Set-Based Logical Problems

★ Find Unique Elements from Multiple Lists

★ Find Common Elements among Three Lists

Remove Duplicates while Preserving Order using Set

★ Check if Two Strings are Anagrams using Set

Find Elements Present in Only One of Two Sets

Count Unique Words in a Sentence

Section 6: Real World Set Applications

★ Unique Visitor Tracker using Set

★ Common Friends Finder (Social Network Problem)

Duplicate Email Detector

★ Tag/Keyword Deduplication System

Set-Based Access Control (Permissions Checker)